

Name: Umex Zaheer

Roll No: 026

Section: A

PROGRAMMING FUNDAMENTAL ASSIGNMENT

3 (PART 4)

B-07-26

Program 1: Sum of All Values in a 2D Array

```
#include <iostream>
using namespace std;
```

```
int main()
```

```
{
```

```
    int rows, columns;
```

```
    cout << "Enter rows and columns: ";
```

```
    cin >> rows >> columns;
```

```
    int numbers[50][50];
```

```
    int sum = 0;
```

```
    cout << "Enter array elements: \n";
```

```
    for (int i = 0; i < rows; i++)
```

```
    {
```

```
        for (int j = 0; j < columns; j++)
```

```
        {
```

```
            cin >> numbers[i][j];
```

```
            sum += numbers[i][j];
```

```
        }
```

```
    }
```



```
cout << "sum of all elements = " << sum;
```

```
return 0;
```

```
}
```

Program 2: Subtract All Values from the first Element

```
#include <iostream>
using namespace std;
```

```
int main()
```

```
{
```

```
    int rows, columns;
```

```
    cin >> rows >> columns;
```

```
    int numbers[50][50];
```

```
    cout << "Enter array elements: \n";
```

```
    for (int i = 0; i < rows; i++)
```

```
    {
```

```
        for (int j = 0; j < columns; j++)
```

```
        {
```



```
    cin >> numbers[i][j];  
    }  
}
```

```
int result = numbers[0][0];
```

```
for(int i = 0; i < rows; i++)  
{  
    for(int j = 0; j < columns; j++)  
    {
```

```
        if (! (i == 0 && j == 0))
```

```
            result = result - numbers[i][j];
```

```
    }  
    }  
cout << "Final Result = " << result;
```

```
    return 0;
```

```
}
```


Program 3: Area Calculator (Function Overloading)

```
#include <iostream>
using namespace std;
```

```
double calculateArea(double side)
```

```
{
```

```
    return side * side;
```

```
}
```

```
double calculateArea(double length, double width)
```

```
{
```

```
    return length * width;
```

```
}
```

```
int main() {
```

```
    double side, length, width;
```

```
    cout << "Enter side of square: ";
```

```
    cin >> side;
```



```
cout << "Enter length and width of rectangle: ";  
cin >> length >> width;
```

```
cout << "Area of Square: " << calculateArea(side)  
    << endl;
```

```
cout << "Area of Rectangle: " << calculateArea  
    (length, width);
```

```
return 0;
```

3

Program 4: Bowling Average Calculator

```
#include <iostream>  
using namespace std;
```

```
double get Bowling Average (int runsConceded, int  
    wicketTaken)
```

```
{
```

```
    if (wicketTaken == 0)
```

```
{
```

```
        return 0.0;
```



```
return (double)runsConceded / wicketsTaken;  
}  
int main() {
```

```
int runsConceded, wicketsTaken;
```

```
cout << "Enter runs conceded and wickets Taken: ";  
cin >> runsConceded >> wicketsTaken;
```

```
cout << " Bowling Average: " << getBowlingAverage  
(runsConceded, wicketsTaken);
```

```
return 0;
```

```
}
```

Program 5: Vertical Word Printer

```
#include <iostream>  
using namespace std;
```

```
void printVertical (char word[], int size)
```

```
{  
    for (int i = 0; i < size; i++)
```



```
cout << "Enter a word: ";  
cin >> word;
```

```
int size = 0;
```

```
while (word[size] != '\0')  
{  
    size++;  
}
```

```
printVertical(word, size);
```

```
return 0;
```

```
}
```

Program 6: Reverse a Word

```
#include <iostream>  
using namespace std;
```

```
void reverseWord(char word[], int size)  
{
```



```
int start = 0;  
int end = size - 1;
```

```
While (start < end)
```

```
{
```

```
    char temp = word[end];  
    word[start] = word[end];  
    word[end] = temp;
```

```
    start++;  
    end--;
```

```
}
```

```
}
```

```
int main() {
```

```
    char word[100];
```

```
    cout << "Enter a word: ";
```

```
    cin cin >> word;
```

```
    int size = 0;
```



```
While (word[size] != '\0')
```

```
{
```

```
    size++;
```

```
}
```

```
reverseWord(word, size);
```

```
cout << "Reversed word: " << word;
```

```
return 0;
```

```
}
```